SUPERSEDING
MIL-D-85465A(AS)
8 October 1986

MILITARY SPECIFICATION

DEVICE, SAFETY AND ARMING, FUZE

FSU-10/A

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the FSU-10/A Safety and Arming Device, Fuze, referred to herein as the fuze.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

UU-T-81	Tag, Shipping and Stock.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1336

AMSC N/A

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BB-N-411	Nitrogen, Technical.

MILITARY

MIL-C-21723	Composition CH-6.
MIL-L-85297	Lead, Explosive Mark 24 Mod 0 Mark 25 Mod 0.

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.
MIL-STD-1695	Environments, Working, Minimum Standards for.
DOD-STD-1866	Soldering Process General (Non-electrical).
MIL-STD-45662	Calibration Systems Requirements.

2.1.2 Other Government drawings and publications. The following other Government drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

1137AS196	Switching Sequence.

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1137AS200	Fuze, Safety and Arming Device, Fuze FSU-10/A.
1137AS284	Explosive Rotor Assembly.
SA1137AS824	Test Chamber, Closed.
1137AS1507	Rotary Switch Assembly.
1137AS1531	Diaphragm.
1137AS1534	Cover Assembly.
1137AS1582	Container, Shipping and Storage.
1137AS2040	Explosive Rotor Nut Assembly.

PUBLICATIONS

Naval Air Systems Command
(Code Ident 30003)

WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

A 108	Steel Bars, Carbon, Cold-Finished, Standard Quality.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171-178 Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item description. The fuze is an electromechanical device which provides two major subfunctions in the AIM-54C Guided Missile. The first subfunction provides safety-arming-firing for the rocket motor, and the second subfunction provides safety-arming for the warhead. The fuze includes a mechanical and electrical out-of-line safety pyrotechnic device which, during the missile launch sequence, arms and initiates the rocket motor. The fuze also includes an explosive-train which is maintained mechanically and electrically out-of-line until certain prerequisite missile launch events have occurred, after which time it arms the warhead. The fuze provides an accurate arming time commensurate with the safe separation time of the missile from the launch aircraft. Upon receipt of a proper firing signal (from the target detecting device (TDD) or the contact fuze triggering device (FTD)), the fuze detonates the booster.

3.2 Characteristics.

3.2.1 Performance. The performance characteristics of the fuze shall be as specified herein.

3.2.1.1 Solenoid actuation.

3.2.1.1.1 Intent-to-launch (ITL) solenoid. The ITL solenoid shall actuate while 97 +23, -27 volts direct current (Vdc) is applied across pin T(+) and c(-) of connector 6J1.

NOTE: The ITL solenoid must operate at 70 Vdc and 120 Vdc and all voltages in between.

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3.2.1.1.2 Active-transfer-initiate (ATI) solenoid. The ATI solenoid shall actuate when a voltage of 97 +23, -27 Vdc is applied across pins T(+) and Y(-) of connector 6J1.

NOTE: The ATI solenoid must operate at 70 Vdc and 120 Vdc and all voltages in between.

3.2.1.2 Ignition safety mechanism (ISM) arming. Mechanical arming of the ISM shall occur within 100 to 250 milliseconds (ms) after actuation of the ITL solenoid and when a force between 110 and 600 pounds is applied along the axis of the pull pin in an extending direction (see Figure 1). The resistance of the armed ISM between pins J(+) and M(-) of connector 9J1 shall be from 2.30 to 3.50 ohms when measured with a resistance measuring device that has a current of not greater than 10 milliampere (mA) while applying 28 ±4 Vdc (fire signal) across pins C(+) and B(-).

3.2.1.3 Resistance, voltage, and capacitance (safe condition). All fuze electrical assemblies shall meet the resistance, voltage, and capacitance requirements specified in Table I at ambient temperature while in the safe condition (see 6.3.1).

3.2.1.4 Resistance (armed condition). With the fuze in the armed condition (see 6.3.2), with and without simulated rocket motor pressure (140 - 600 pounds) applied, the resistance between pins L(+) and c(-) of connector 6J1 shall be 1.5 to 4.6 ohms.

3.2.1.5 Electrical arming sequence. The voltage steps recorded across pins P(+) and c(-) of connector 6J1 shall be in accordance with the sequence shown on Figure 2 when the input voltage at pins T(+) and b(-) is 97 +23, -27 Vdc and the fuze is operated through the normal functional cycle. Also the voltage pulse recorded across pins R(+) and c(-) of connector 6J1 shall be in accordance with Figure 3.

3.2.1.6 Safe-arm (S-A) arming.

3.2.1.6.1 Late ATI. After ISM arming, the warhead S-A assembly shall run to and stop at the ATI stop position when 350 to 900 pounds per square inch (psi) simulated rocket motor pressure is applied to the fuze. The fuze shall arm upon ATI solenoid actuation.

3.2.1.6.2 Early ATI. The fuze shall arm within 8.0 to 11.0 seconds, and switching shall be as shown on Figure 2.

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3.2.1.6.3 ITL solenoid deactivation. When the S1D switch opens at 35 degrees rotation during S-A arming sequence, the ITL solenoid shall deactivate in less than 1.4 seconds after this occurance and return to the unenergized state locking the arming energy into the fuze S-A assembly.

3.2.1.7 Safety determination. The fuze shall be in the safe condition (see 6.3.1) prior to, remain safe during, and no explosive or pyrotechnic component shall initiate as a result of the following test sequence:

a. Application of 50 ±2 Vdc across pins L(+) and c(-) of connector 6J1 for not less than 100 ms.

b. Application of 28 ±4 Vdc across pins J(+) and M(-) of connector 9J1 while applying 28 ±4 Vdc across pins C(+) and B(-) of connector 9J1 for not less than 100 ms.

3.2.1.8 Seal. The leakage rate of the fuze shall be in accordance with Drawing 1137AS200.

3.2.2 Environmental conditions. The fuze shall not be damaged nor shall safety or subsequent performance be degraded by exposure to the environments specified in 3.2.2.1 through 3.2.2.5.

3.2.2.1 Temperature and humdity. The fuze shall be safe (see 6.3.3) and operable (see 6.3.4) after exposure to the environment specified in MIL-STD-331, Test 105.1, Single Chamber-Cycling Method.

3.2.2.2 Transportation vibration-temperature. The fuze shall be safe (see 6.3.3) and operable (see 6.3.4) after exposure to the environment specified in MIL-STD-331, Test 119, Procedure 1.

3.2.2.3 Captive carry flight vibration - temperature. During captive carry flight the fuze shall remain safe (see 6.3.3) and operable (see 6.3.4) following exposure to random vibration of 0.01 gravity unit squared per hertz (g^2/Hz) from 20 to 500 Hz and 0.12 g^2/Hz from 1000 to 2000 Hz at temperatures of -65 degrees Fahrenheit (°F), ambient, or +160°F for 2 hours along each of three mutually perpendicular axes (see Figure 4).

3.2.2.4 Missile flight vibration at ATI stop position. The fuze shall remain in the ATI stop position during exposure to not less than 15 minutes of random vibration (5 minutes per axis) having a spectral density of 0.02 g^2/Hz, over a bandwidth of 20 to 2000 Hz, with a roll-off of not less than 6 dB per octave (see Figure 5). After exposure to the environment, the fuze shall move to the armed position upon ATI solenoid actuation. The ISM shall remain in the armed condition throughout this testing.

3.2.2.5 Pull pin safety. The pull pin shall not extend greater than 0.150 inch and the fuze (ISM and S-A assemblies) shall remain safe when a force of 900 ±10 pounds is applied to the pull pin in an extending direction (see Figure 1).

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3.2.3 Environmental safety.

3.2.3.1 Twelve meter drop. The fuze shall be safe (see 6.3.3) to handle and dispose of, but need not be operable (see 6.3.4), after exposure to the environment specified in MIL-STD-331, Test 103.2, Procedure I. No explosive or pyrotechnic component (see MIL-L-85297) shall have burned or detonated during exposure.

3.2.3.2 Jolt. The fuze shall be safe (see 6.3.3) to handle, store, and transport, but need not be operable (see 6.3.4), after exposure to the environment specified in MIL-STD-331, Test 101.3, Procedure I. No explosive or pyrotechnic component (see MIL-L-85297) shall have detonated or burned during exposure.

3.2.4 Explosive train.

3.2.4.1 Static detonator safety. There shall be no evidence of detonation, burning, charring, scorching, melting, or rupturing of any of the explosive leads when the detonators are initiated with the arming rotor in the safe position (see 6.3.1). Smudging of the surfaces of the explosive leads, due to deposits resulting from initiation of the detonators, shall be acceptable. The fuze housing shall not rupture.

3.2.4.2 Explosive train output. The explosive train of the fuze shall produce an average depth of dent of not less than 0.035 inch when armed and initiated on a steel dent block (see Figure 6).

3.2.5 Pyrotechnic train.

3.2.5.1 Static squib safety. There shall be no evidence of squib efflux leakage outside the fuze housing, nor shall any external diaphragm (Drawing 1137AS1531) rupture when the squibs are initiated with the ISM (mechanical barrier) in the safe position (see 6.3.1).

3.2.5.2 Pyrotechnic train output. The squibs shall function and produce a pressure in the closed test chamber (see Drawing SA1137AS824) of not less than 250 pounds per square inch gage (psig) in 9 ms or less when 28 ±4 Vdc is applied across pins J(+) and M(-) of connector 9J1 and 28 ±4 Vdc is momentarily (not less than 50 ms or greater than 5 seconds) applied to pins C(+) and B(-) of connector 9J1.

3.2.6 Visual. The fuze shall meet the following requirements:

a. There shall be no external damage, such as cracks, broken windows, dents, or other defects, which may interfere with functioning or fit of the fuze at the next assembly.

b. A band of yellow adhesive tape shall be present around the fuze mechanism body as shown on Drawing 1137AS200.

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c. Both the ISM and S-A assembly shall be in the safe position (see 6.3.1), as evidenced by a white letter "S" on a green background that is fully visible through the windows located in both ends of the fuze.

d. The safe condition tag in accordance with Figure 7 shall be properly signed by both the contractor and Government inspectors. The safe condition tag shall be securely wired to the fuze using a lead and wire seal to indicate that a successful safety test has been performed. The location of the tag and routing of wire, as well as the lead and wire seal, shall be in accordance with Drawing 1137AS200.

3.3 First article. Unless otherwise specified in the contract or purchase order, the contractor shall furnish 30 fuzes for first article inspection and approval (see 4.2 and 6.2.1). The first article shall be manufactured using the same methods, materials, processes, and procedures proposed for production. Any production prior to acceptance of the first article shall be at the risk of the contractor.

3.4 Design and construction.

3.4.1 Production drawings. The fuze shall meet all the requirements of and be manufactured and assembled in accordance with Drawing 1137AS200.

3.4.2 Standards of manufacture.

3.4.2.1 Working environment. The working environment shall meet the minimum levels and requirements of MIL-STD-1695.

3.4.2.2 Soldering. Procedures and requirements for preparation and soldering of electrical connections shall be in accordance with WS-6536. Mechanical soldering shall be in accordance with DOD-STD-1866.

3.4.2.3 Workmanship. The fuze, including all parts and accessories, shall be manufactured in accordance with the applicable drawings and documents specified herein. Workmanship shall meet the requirements of MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein (see 6.2.1). Except as otherwise specified in the contract or purchase order, the contractor

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may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the contracting activity (see 6.2.1). The contracting activity reserves the right to perform or witness any of the inspections set forth in this specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the contracting activity for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the contracting activity to acceptance of defective material.

4.1.1.1 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspections (see 4.2).

b. Quality conformance inspections (see 4.3).

(1) Drawing conformance (see 4.3.1).

(2) Individual acceptance tests (see 4.3.2).

(3) Lot acceptance tests (LAT) (see 4.3.3).

4.2 First article inspections. Unless otherwise specified in the contract or purchase order, first article inspections shall be performed by the contractor and an independent test facility.

4.2.1 First article inspection sequence. The first article fuzes shall meet the test requirements of drawing conformance (4.3.1) and the individual acceptance tests (4.3.2) of the quality conformance inspections. The first article fuzes shall then be subjected to the inspections shown on Figure 8. Inspections shown on Figure 8 shall be performed at the independent test facility designated in the contract or purchase order (see 6.2.1). Unless otherwise specified in the contract or purchase order, first article fuzes shall be shipped to the contracting activity after receipt of first article approval (see 6.2.1).

4.2.2 First article acceptance criteria. The acceptance criteria for first article approval shall be the successful completion of the testing requirements shown on Figure 8.

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4.3 Quality conformance inspection.

4.3.1 Drawing conformance. The quality system specified in the contract or purchase order shall assure that the fuzes are manufactured and assembled in accordance with the requirements of 3.4.1.

4.3.2 Individual acceptance tests. Each fuze shall meet the inspections and tests shown on Figure 9 in the sequence shown.

4.3.3 Lot acceptance test.

4.3.3.1 Serialization. As specified in the contract or purchase order (see 6.2.1), each fuze successfully meeting the requirements of the individual acceptance test of 4.3.2 shall be assigned a Government serial number. After assignment of a Government serial number the fuzes shall be formed into an inspection lot as specified herein.

4.3.3.2 Inspection lot size. Unless otherwise specified in the contract or purchase order (see 6.2.1), an inspection lot shall contain not less than 51 nor greater than 500 fuzes, including those required for destructive tests. An inspection lot, as defined in MIL-STD-105, shall consist of completed fuzes fabricated under the same conditions and offered for inspection at the same time.

4.3.3.3 Lot acceptance test. Unless otherwise specified in the contract or purchase order, 15 fuzes shall be randomly selected from the inspection lot by the contracting activity and subjected to the environmental tests and inspections shown on Figure 10 at an independent testing facility. The remainder of the lot shall be held in bond pending successful completion of the LAT.

4.3.3.4 Inspection lot acceptance criteria. The criteria for accepting the inspection lot is the successful completion of the LAT.

4.4 Test equipment and provisions.

4.4.1 Test equipment. The contractor shall provide and maintain an adequate system of inspection and test equipment necessary to ensure that parts and products will meet the contract or purchase order, specification, and drawing requirements. Unless otherwise specified herein, the magnitude of any error introduced by the test equipment shall be not greater than 10 percent of the allowable tolerance of the requirement being measured. All test equipment utilized by the contractor for performance of the inspections specified herein shall meet the system requirements specified in MIL-STD-45662.

4.4.2 Test equipment calibration. When specified in the contract or purchase order (see 6.2.2), a calibration plan shall be prepared and shall be approved by the contracting activity prior to use.

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4.4.3 Test environment. Unless otherwise specified herein, the inspection methods of 4.5 shall be performed under the conditions specified in 3.4.2.1.

4.4.4 Acceptance test plan (ATP). When specified in the contract or purchase order (see 6.2.2), the contractor shall prepare an ATP for approval by the contracting activity. The ATP shall be approved prior to any acceptance testing.

4.4.5 Fuze inspection record. When specified in the contract or purchase order (see 6.2.2), an inspection record shall be completed and delivered for each fuze. The inspection record format shall be acceptable to the contracting activity and shall contain a minimum of information as shown on Figure 11.

4.4.6 Simulated rocket motor pressure. Rocket motor pressure shall be simulated by a dry nitrogen source meeting the requirements of BB-N-411, Class 1, Grade A.

4.5 Methods of inspection. The inspections of 4.5.1 through 4.5.14 shall be performed in the sequence shown on Figures 8 through 10, to determine conformance to the applicable requirements of Sections 3 and 5.

4.5.1 Solenoid actuation.

4.5.1.1 ITL solenoid. To determine conformance to 3.2.1.1.1, the ITL solenoid shall be tested as follows:

a. Connect a voltmeter, having an input impedance of not less than 10 megohms (Mohms), across pins P(+) and c(-) of connector 6J1.

b. Apply 70 +2, -0 Vdc across pins T(+)and c(-) of connector 6J1. (After shipment to the independent testing facility the voltage applied across pins T(+) and c(-) of connector 6J1 shall be 85 ±2 Vdc.)

c. Apply 28 ±4 Vdc across pins J(+) and M(-) of connector 9J1.

d. Momentarily (not less than 50 ms or greater than 5 seconds) apply 28 ±4 Vdc across pins E(+) and D(-) of connector 9J1.

e. A voltage of 1.6 to 2.4, for 70 Vdc input, or 1.9 to 2.8 Vdc for 85 Vdc input, measured across pins P(+) and c(-) of connector 6J1 will indicate operation of the ITL solenoid and turn on of photo transistor Q4.

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4.5.1.2 ATI solenoid. To determine conformance to 3.2.1.1.2, the ATI solenoid shall be tested as follows:

a. Connect a voltmeter, having an input impedance of not less than 10 Mohms, across pins P(+) and c(-) of connector 6J1.

b. Apply 70 +2, -0 Vdc across pins T(+) and c(-) of connector 6J1. (After shipment to the independent testing facility the voltage applied across pins T(+) and c(-) of connector 6J1 shall be 85 ±2 Vdc.)

c. Apply 28 ±4 Vdc across pins J(+) and M(-) of connector 9J1.

d. Apply 70 +2, -0 Vdc across T(+) and Y(-) of connector 6J1. (After shipment to the independent testing facility the voltage applied across T(+) and Y(-) of connector 6J1 shall be 85 ±2 Vdc.)

e. A voltage of 2.4 to 3.2 Vdc, for 70 Vdc input, or 2.9 to 3.9 Vdc, for 85 Vdc input, measured across P(+) and c(-) of connector 6J1 will indicate operation of the ATI solenoid.

4.5.2 ISM arming. To determine conformance with 3.2.1.2, the electrical and mechanical arming of the ISM shall be tested as follows:

WARNING

All tests of the fuze where arming occurs shall be performed with the fuze installed behind an explosive-proof barrier.

a. Enable the ITL solenoid by applying 70 +2, -0 Vdc across pins T(+) and c(-) of connector 6J1. After shipment to the independent testing facility enable the ITL solenoid by applying 85 ±2 Vdc across pins T(+) and c(-) of connector 6J1.

b. Apply 28 ±4 Vdc across pins J(+) and M(-) of connector 9J1.

c. Actuate the ITL solenoid circuit by momentarily (not less than 50 ms or greater than 5 seconds) applying 28 ±4 Vdc across pins E(+) and D(-) of connector 9J1.

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d. Determine electrical and mechanical arming of the ISM by measuring the time between first motion of the pull pin after the detent is overcome and the opening of switch S6A. Opening of switch S6A will be indicated by a drop in voltage across pins P(+) and c(-) of connector 6J1 (see Figure 2).

e. Mechanical arming shall be accomplished by applying a force of 355 ±245 pounds to the pull pin in an extending direction (see Figure 1). The test equipment used to perform pull pin extension shall be capable of fully extending the pull pin at a rate of not less than 30 inches per second.

f. Electrical and mechanical arming of the ISM shall occur within 175 ±75 ms after first motion of the pull-pin and the voltage levels recorded at pins P(+) and c(-) of connector 6J1 shall be 1.05 ±0.25 Vdc for 70 +2, -0 Vdc input or 1.3 ±0.3 for 85 ±2 Vdc input.

g. Remove the voltages applied in steps a, b, and c above.

WARNING

For the following test, application of more than the specified current to pins J(+) and M(-) of connector 9J1 may result in firing of the squibs.

h. Connect a resistance measurement device with a current not greater than 10 mA between pins J(+) and M(-) of connector 9J1. Apply 28 ±4 Vdc across pins C(+) and B(-) of connector 9J1. The resistance between pins J(+) and M(-) of connector 9J1 shall conform to 3.2.1.2.

4.5.3 Resistance, voltage, and capacitance (safe condition).

4.5.3.1 Resistance and voltage (safe condition). To determine conformance to 3.2.1.3 the resistance between and the voltage across the test points of Table I, Items 1 through 26 shall be measured with the fuze at ambient temperature in the safe condition (see 6.3.1). The following conditions shall apply:

a. For Item 1, the megohmmeter used shall have not more than 100 microamperes of test current. For measurement of Item 1 a test voltage of 500 ±40 Vdc shall be used.

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b. For measurement of Items 4 and 22, enable the ITL solenoid circuit by momentarily (not less than 50 ms or greater than 5 seconds) applying 28 ±4 Vdc across pins E(+) and D(-) of connector 9J1. For measurement of Item 5, disable the ITL solenoid circuit by momentarily (not less than 50 ms or more than 5 seconds) applying 28 ±4 Vdc across pins H(+) and D(-) of connector 9J1.

c. For measurement of Items 10 through 21, subtract any resistance contributed by the test setup, including the test connector, from the test results.

NOTE: After the successful completion of Items 10 through 20 (common wire test) any common wire may be used in place of the specified wire in the remainder of tests in Section 4.

CAUTION

During measurement of Items 22, 23, and 24, to prevent damage to the solenoid coil, the actuation voltage (100 ±5 Vdc) shall not be applied for greater than 20 seconds at a time or exceed an average of 20 seconds in a 140 second period (14 percent duty cycle).

d. For measurement of Items 22 and 23, apply 100 ±5 Vdc through a current-limiting resistor of 2.0 ±0.1 kilohms (kohms), 5 watts (W), or dynamically limit current to 50 mA, across pins Z(+) and T(-) of connector 6J1 for Item 22 and across pins Y(+) and T(-) of connector 6J1 for Item 23. Connect a voltmeter across pins Z(+) and T(-), and across pins Y(+) and T(-) of connector 6J1 to measure the forward voltage of diodes CR9 and CR7 respectively.

e. For measurement of Item 24, apply 100 ±5 Vdc through a current-limiting resistor of 2.2 ±0.1 kohms, 2 W, or dynamically limit current to 50 mA, to pins C(+) and c(-) of connector 6J1. Connect a voltmeter across pins C(+) and c(-) to measure the zener voltage (V_Z) of VR10.

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f. For measurement of Item 25, apply 100 ±5 Vdc to pins C(+) and c(-) of connector 6J1. Connect a voltmeter across pins A(+) and b(-) of 6J1 to measure the voltage across R2.

4.5.3.2 Capacitance (safe condition). To determine conformance to 3.2.1.3, the capacitance between the test points specified in Table I, Item 26, shall be measured with a capacitance bridge or equal test method, with the fuze in the safe condition (see 6.3.1).

4.5.4 Resistance (armed condition).

WARNING

All tests of the fuze in the armed condition (see 6.3.2) shall be performed with the fuze installed in a safety chamber capable of containing the explosive output of 3.5 grams of Composition CH-6 (see MIL-C-21723).

WARNING

Test points L to c are across the detonators. Test current in excess of 5 mA may cause initiation of explosive components.

To determine conformance to 3.2.1.4, the resistance between pins L(+) and c(-) of connector 6J1 shall be measured with the fuze in the armed condition (see 6.3.2). The test equipment used shall limit the test current to not greater than 5 mA . The procedure of 4.5.3.1c shall apply.

4.5.5 Electrical arming sequence. To determine conformance to 3.2.1.5, the voltages shall be measured as follows:

a. Connect the fuze case and pin L of connector 6J1 to pin c of connector 6J1.

b. Record the voltages across pins P(+) and c(-) (Developmental TM), and pins R(+) and c(-) (Fuze Tactical TM) all of connector 6J1.

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WARNING

All tests of the fuze in the armed condition (see 6.3.2) shall be performed with the fuze installed in a safety chamber capable of containing the explosive output of 3.5 grams of Composition CH-6 (see MIL-C-21723).

c. Arm the fuze as specified in 4.5.5.1 and 4.5.5.2.

d. Use the voltage/time record to determine whether the voltage steps measured at pins P and R conform to 3.2.1.5.

e. At the completion of all arming test sequences, the fuze shall be recycled to the safe position (see 6.3.1) as specified in 4.5.6.1.

NOTE: Although failure to recycle to the safe position does not affect acceptance of a lot, any fuze which cannot be recycled shall be rejected (see 4.5.6.2).

4.5.5.1 S-A arming time. For arming tests that require the measurement of S-A arming time, the following description and conditions shall apply:

a. Arming time is considered as the period of time starting with the first movement of the explosive rotor assembly and ending when the warhead explosive train is fully aligned.

b. Determination of the S-A arming time shall be made in the following manner. Using the voltage/time record, 4.5.5b and d, measured at pin P of connector 6J1, measure the time span between 15 and 35 degrees of rotation of the explosive rotor assembly (T_1). Next measure the time span between 10 and 77 degrees of rotation of the S-A rotary switch assembly (T_2). Calculate the S-A arming time (T_A) by the following equation: (see Figure 12).

$$T_A = 1/2\ T_1 + T_2$$

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4.5.5.2 Late ATI arming. To determine conformance to 3.2.1.6.1, the warhead S-A assembly shall be tested as follows:

a. With the ISM in the armed condition (see 4.5.2), apply 70 +2, -0 Vdc across pins T(+) and c(-) of connector 6J1. After shipment to the independent testing facility, apply 85 ±2 Vdc across pins T(+) and c(-) of connector 6J1.

b. Apply 28 ±4 Vdc across pins J(+) and M(-) of connector 9J1.

c. Apply a pressure of 625 ±275 psig to the fuze pressure piston. The pressure shall be fully applied within 140 ms.

d. Measure the voltage across pins P(+) and c(-) of connector 6J1 using a voltage recorder, 4.5.5b and d, which has an input impedance of not less than 10 Mohms while the explosive rotor assembly rotates from the safe position to the ATI stop position. The voltages recorded shall conform to 3.2.1.6.1.

e. While monitoring the voltage across pins P(+) and c(-) of connector 6J1, actuate the ATI solenoid by connecting pin J to pin c of connector 6J1. The voltage across pins P and c shall change from not greater than 0.40 Vdc to not less than 1.50 Vdc, and back to not greater than 0.40 Vdc (S-A armed).

4.5.5.3 Early ATI arming. To determine conformance to 3.2.1.6.2, the fuze shall be tested as follows:

a. Repeat 4.5.5.2a through c except 120 +0 -2 Vdc shall be applied across pins T(+) and c(-) of connector 6J1. At the independent testing facility, 115 ±2 Vdc shall be applied across Pins T(+) and c(-) of connector 6J1.

b. Measure the voltage across pins P(+) and c(-) of connector 6J1 using a voltage recorder which has an input impedance of not less than 10 Mohms while the explosive rotor assembly rotates from the safe position to the full armed position. The voltages recorded shall conform to 3.2.1.6.2.

c. Within 4.5 seconds of application of pressure to the fuze pressure piston (see 4.5.5.2c), actuate the ATI solenoid by connecting pin J to pin c of connector 6J1.

d. Measure the arming time (see 4.5.5.1). Arming time shall conform to 3.2.1.6.2.

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4.5.5.4 ITL solenoid deactivation. To determine conformance to 3.2.1.6.3, the voltage trace of 4.5.5.3 shall be used as follows:

a. Measure from the voltage transition of the last opening of S1D (see Figure 2) to voltage transition of ITL deactiviation.

b. The measured time shall conform to 3.2.1.6.3.

4.5.6 Safety determination. To determine conformance to 3.2.1.7 the fuze shall be tested as follows:

a. If required, recycle the fuze to the safe position (see 6.3.1) in accordance with 4.5.6.1. Fuzes that fail to recycle shall be handled in accordance with 4.5.6.2.

b. Visually verify that the fuze is in the safe position (see 6.3.1), as evidenced by a white letter "S" on a green background that is fully visable through the windows located in both ends of the fuze.

WARNING

Do not proceed with the test unless the fuze is in the safe position (see 6.3.1). The test of c below will cause initiation of the explosive train or the pyrotechnic train if the fuze is not in the safe position.

c. Place the fuze in a safety chamber capable of containing an explosive charge generated by 3.5 grams of Composition CH-6. Apply 50 ±2 Vdc across pins L(+) and c(-) of connector 6J1 (detonator circuit). Apply 28 ±4 Vdc across pin J(+) and K(-) of connector 9J1. For not less than 100 ms or greater than 1 second apply 28 ±4 Vdc across pins C(+) and B(-), and across pins N(+) and P(-) of connector 9J1 (squib circuit).

d. Remove the voltages applied in step c above.

e. Remove the fuze from the safety chamber. Visually verify that the fuze is in the safe position (see 6.3.1) as specified in b above and conforms to 3.2.6c.

f. Fill in, sign, and attach a safe condition tag (see Figure 7). Immediately attach a lead and wire seal (see 3.2.6d) to indicate that safety determination has been made.

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4.5.6.1 Procedure for recycling from armed to safe position. Fuzes in the armed position (see 6.3.2) shall be recycled to the safe position (see 6.3.1) as follows:

a. Apply a pressure of 625 ±275 psig to the fuze pressure piston.

CAUTION

To avoid fuze damage, the voltage of b below shall be applied for not greater than 1 minute at a time, or exceed an average of 1 minute in a 7 minute period (14 percent duty cycle).

b. Momentarily (not less than 50 ms or greater than 5 seconds) apply 28 ±4 Vdc across pins E(+) and D(-) of connector 9J1.

c. Apply and maintain 100 ±5 Vdc across pins T(+) and Z(-), and across pins T(+) and Y(-) of connector 6J1 for not greater than 1 minute.

d. Remove pressure from fuze pressure piston until 150 psig is reached. Below 150 psig, gradually (30 psig per second) remove pressure from the fuze pressure piston. Do not go to next step (e) until S-A assembly has resafed.

e. Fully depress the pull-pin at a rate not greater than 0.5 inch per second until the pull-pin is returned to the fully retracted position.

f. Momentarily (not less than 50 ms or greater than 5 seconds) apply 28 ±4 Vdc across pins H(+) and D(-) of connector 9J1.

g. Remove all electrical inputs to the fuze.

h. Visually observe that the fuze is in the safe position (see 6.3.1) as specified in 4.5.6b. If the fuze is not in the safe position, proceed in accordance with 4.5.6.2.

4.5.6.2 Handling of fuzes that fail to recycle.

WARNING

Proper safety precautions shall be observed when handling armed fuzes which fail to recycle to the safe position (see 6.4).

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Unless otherwise specified in the contract or purchase order (see 6.2.1), disposition of armed fuzes that repeatedly fail to recycle to the safe position (see 6.3.1) shall be at the discretion of the cognizant demolition officer. The unsafe condition shall be indicated by securely attaching an unsafe condition tag as shown on Figure 13.

4.5.7 Seal tests. To determine conformance to 3.2.1.8, the fuze shall be tested as follows. Using test equipment capable of measuring helium leakage at a rate not greater than 1×10^{-7} standard atmosphere cubic centimeters per second (std atm cm^3/s), the fuze shall meet the leakage rate requirements as shown on Drawing 1137AS200.

4.5.8 Environmental tests. Fuzes of the first article and a randomly selected LAT sample of 15 fuzes from an inspection lot shall be subjected to the following environmental tests as shown on Figure 8 or 10. Any fuze subjected to these tests and the tests of 4.5.9 shall be considered destroyed and shall not be returned to the lot nor forwarded for service use, but shall be labeled appropriately and held for disposal as specified in the contract or purchase order (see 6.2.1). During any test, initiation of any explosive component, except those which are initiated intentionally as part of a test, shall be considered a critical defect and shall be cause for rejection of the lot from which the fuze was drawn. Throughout the environmental tests, the fuze shall be in the safe condition (see 6.3.1) unless otherwise specified herein.

4.5.8.1 Temperature and humidity. To determine conformance to 3.2.2.1, the unarmed fuze shall be tested in accordance with MIL-STD-331, Test 105.1, Single Chamber-Cycling Method.

NOTE: Thoroughly dry the electrical connectors prior to performing subsequent tests.

4.5.8.2 Transportation vibration-temperature. To determine conformance to 3.2.2.2, the unarmed fuze shall be tested in accordance with MIL-STD-331, Test 119, Procedure 1.

4.5.8.3 Captive carry flight vibration-temperature. To determine conformance to 3.2.2.3, the fuze shall be subjected to random vibration of 0.01 g^2/Hz from 20 to 500 Hz and 0.12 g^2/Hz from 1000 to 2000 Hz at -65°F, ambient, or +160°F for two hours along each of three mutually perpendicular axes. (See Figure 4). Test in accordance with MIL-STD-810, Method 514.3 except to previously defined levels.

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NOTE: If required to protect test equipment, vibration amplitude peaks may be limited to three times the average amplitude.

4.5.8.4 Missile flight vibration at ATI stop position. To determine conformance to 3.2.2.4, the fuze shall be subjected to a total of not less than 15 minutes (5 minutes per axis) of random vibration having a spectral density of 0.02 g^2/Hz, over a bandwidth from 20 to 2000 Hz, with a roll-off of not less than 6 dB per octave (see Figure 5). The fuze shall be tested in accordance with MIL-STD-810, Method 514.3 except to the previously defined levels. The test sequence shall be as follows:

a. Position the S-A assembly in the ATI stop position by performing 4.5.5.2a through d.

b. Apply vibration for not less than 5 minutes along each of the three mutually perpendicular axes (vertical, horizontal, and longitudinal). The S-A assembly shall remain at the ATI stop position during exposure to vibration, and the ISM shall remain in the armed position.

c. Upon completion of vibration, snap arm the S-A assembly from the ATI stop position to the armed position by performing 4.5.5.2e.

d. Visually verify that the fuze is in the armed position (see 6.3.2) as evidenced by a white letter "A" on a red background clearly visible through the window located on the warhead S-A end of the fuze.

e. The fuze shall be recycled to the safe position (see 6.3.1) as specified in 4.5.6.1.

4.5.8.5 Pull pin safety. To determine conformance to 3.2.2.5 apply a force of 900 ±10 pounds to the pull pin in an extending direction. The force shall be fully applied within 10 seconds and shall remain applied for 12 ±2 seconds.

4.5.9 Environmental safety test.

4.5.9.1 Twelve meter drop. To determine conformance to 3.2.3.1, the fuze, which has been installed in a test fixture simulating an inert rocket motor and attached to a simulated inert armament section, shall be tested in accordance with MIL-STD-331, Test 103.2, Procedure I except that the drop orientations shall be as follows (see Figure 1):

a. Longitudinal axis of the test vehicle horizontal.

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b. Longitudinal axis of the test vehicle horizontal, transverse to the axis of a above.

c. Longitudinal axis of the test vehicle vertical, with the S-A end of the fuze up.

4.5.9.2 Jolt. To determine conformance to 3.2.3.2, the fuze shall be tested in accordance with MIL-STD-331, Test 101.3, Procedure I.

4.5.10 Explosive-train tests.

4.5.10.1 Static detonator safety. To determine conformance to 3.2.4.1 and to accomplish detonator initiation with the fuze in the safe condition (see 6.3.1), the fuze shall be modified and tested as follows:

WARNING

Proper safety precautions must be observed when performing the following modifications (see 6.4). Care must be taken to prevent accidental initiation of the detonators by static potentials. The fuze shall be placed behind an explosive - proof barrier.

a. Verify that the fuze meets the safe condition requirements specified in 4.5.6a and b.

b. Remove the fuze cover assembly (see Drawing 1137AS1534).

c. Remove the explosive rotor nut assembly (see Drawing 1137AS2040) and the rotary switch assembly (see Drawing 1137AS1507) from the explosive rotor assembly (see Drawing 1137AS284).

d. Tin the ends of a 2 inch long piece of #4 copper braid.

e. Tin ajacent sections of the safe-arm stator switch assembly traces 1 and 12, also 4 and 5 (see Figure 14).

f. Solder one end of the braid of d above to safe-arm stator switch assembly trace number 1 and solder across to safe-arm stator switch assembly trace number 12 (see Figure 14) and cut off the excess.

g. Repeat step f using traces 5 and 4.

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h. Replace rotary switch assembly, explosive rotor nut, and cover assembly.

i. Attach an electrical cable, containing conductors to pins M and W of connector 6J1, to the fuze.

j. Initiate the detonators by applying 100 ±5 Vdc, with a capability of not less than 1 ampere, across the conductors attached to pins M(+) and W(-) of connector 6J1.

4.5.10.2 Explosive train output. The explosive train output test shall consist of initiating the fuze on a steel dent block and measuring the depth of the dent to determine the output. To determine conformance to 3.2.4.2, the fuze shall be tested as follows and the depth of the dent shall be measured as specified in 4.5.10.2.1:

WARNING

The explosive train output test involves hazardous operations and requires the exercise of extremely stringent safety precautions and procedures. Suitable equipment, facilities, and comprehensive safety reviews must be utilized (see 6.4).

WARNING

Proper safety precautions must be observed when making the following modifications and care must be taken to prevent accidental initiation of the detonators (see 6.4).

a. Verify that the fuze meets the safe condition requirements specified in 4.5.6a and b.

b. Cycle the fuze to the ATI stop position in accordance with 4.5.5.2a through d.

c. Remove the inputs specified in 4.5.5.2a through d. This will result in the storage of arming energy for later use.

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d. Place the fuze on the dent block shown on Figure 6 so the output end of the S-A cover assembly is 0.325 ±0.025 inch from the dent block.

e. Attach an electrical cable containing conductors to pins T, C, c, and J of connector 6J1.

f. Place the fuze with the cable and dent block behind a safety barrier.

g. Arm the fuze by applying and maintaining 100 ±5 Vdc, with a current capability of not less than 1 ampere, across pins T(+), c(-), and J(-).

h. Initiate the detonators by shorting across pins C and c.

4.5.10.2.1 Output dent measurement procedure. The output dent shall be measured as follows:

a. Remove all foreign material from the dent block.

b. Place the gage probe of a dial-indicator height gage in the deepest part of the dent and zero the gage indicator (see Figure 6).

c. Remove the probe from the dent and take gage readings at two points, each 0.125 ±0.030 inch from the opposite edge of the dent block as shown on Figure 6. Average the two readings to determine the average depth of dent and conformance to 3.2.4.2.

4.5.11 Pyrotechnic train tests.

4.5.11.1 Static squib safety. To determine conformance to 3.2.5.1, the fuze shall be tested as follows:

WARNING

Proper safety precautions shall be observed when conducting the following test (see 6.4). Care shall be taken to prevent accidental initiation of the squibs.

a. Verify that the fuze meets the safe condition requirements specified in 4.5.6a and b.

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b. Apply 100 ±5 Vdc across pins T(+) and c(-) of connector 6J1. Momentarily (not less than 50 ms or greater than 5 seconds) apply 28 ±4 Vdc across pins E(+) and D(-) of connector 9J1. Apply 28 ±4 Vdc across pins J(+) and M(-) of connector 9J1.

c. Connect a voltmeter, having an input impedance of not less than 10 Mohms to pins P(+) and c(-) of connector 6J1 and monitor the voltage as the pull pin is very slowly extended (see Figure 1). Extend the pull pin only to the point where the voltage across pins P and c drops from greater than 2.6 Vdc to less than 1.7 Vdc. Do not extend the pull pin past this point.

d. Remove all voltage inputs applied to the fuze in b above.

e. Attach an electrical cable, containing conductors to pins B, C, J, and M of connector 9J1 to the fuze.

f. Place the fuze, with attached electrical cable, behind an explosive-proof barrier.

g. Apply 28 ±4 Vdc, with a current capability of not less than 10 amperes for 30 ms, to the conductors attached to pins J(+) and M(-) of connector 9J1.

h. Initiate the squibs by momentarily (not less than 50 ms or greater than 5 seconds) applying 28 ±4 Vdc to the conductors attached to pins C(+) and B(-) of connector 9J1.

4.5.11.2 Pyrotechnic train output. To determine conformance to 3.2.5.2, the output of the squibs shall be tested as follows:

a. Attach the closed test chamber shown on Drawing SA1137AS824 and appropriate instrumentation to the ISM end of the fuze.

b. Arm the ISM using the procedures of 4.5.2a through e.

c. Apply 28 ±4 Vdc, with a current capability of not less than 10 amperes for 30 ms, across pins J(+) and M(-) of connector 9J1 (squib circuit).

d. Momentarily (not less than 50 ms or greater than 5 seconds) apply 28 ±4 Vdc across pins C(+) and B(-) of connector 9J1 (K2 relay circuit).

e. Measure and record the resultant pressure versus time output of the squibs.

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4.5.12 Drawing conformance inspections. To determine conformance to 3.4.1, drawing conformance inspections shall be performed on parts, subassemblies, and assemblies of the fuze, and on each completely assembled, sealed fuze in accordance with Drawing 1137AS200 and all lower tier drawings.

4.5.13 Visual inspection. The fuze shall be visually inspected to determine conformance to 3.2.6. The safety determination tag and the lead wire seal, 3.2.6d, shall only be verified after the final individual acceptance tests of Figure 9.

4.5.14 Packaging inspection. Prior to shipment, the packaging, packing, and marking of the fuze shall be inspected to determine conformance to Section 5.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with FED-STD-102, Level C, and as specified on Drawing 1137AS1582.

5.1.1 Level C packaging. Fuzes shall be packaged in accordance with Drawing 1137AS1582 and in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during direct shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be in accordance with FED-STD-102, Level C, and as specified on Drawing 1137AS1582.

5.2.1 Level C packing. Three fuzes, packaged in accordance with 5.1.1, shall be packed in each shipping container in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. Packing and containers shall be suitable for explosives and shall conform to 49 CFR 171-178.

5.3 Marking. In addition to any special marking required by the contract or purchase order, marking shall be in accordance with Drawing 1137AS1582 and MIL-STD-129 (see 6.2.1).

6. NOTES

6.1 Intended use. The fuze is intended for use in the Phoenix AIM-54C Guided Missile.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Number, title, and date of this specification.

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b. First article, if required (see 3.3).

c. Responsibility for inspection and inspection facilities, if other than as specified in 4.1 and 4.2.

d. Disposition of first article samples if other than as specified in 4.2.1.

e. Independent test facility to perform the inspections shown on Figures 8 and 10 (see 4.2.1).

f. Assignment of Government serial numbers (see 4.3.3.1).

g. Inspection lot size if other than as specified in 4.3.3.2.

h. Disposition of unsafe fuzes, if other than as specified in 4.5.6.2.

i. Disposal of fuzes subjected to environmental tests (see 4.5.8).

j. Special marking, if required (see 5.3).

6.2.2 Data requirement. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below should be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data Requirements Title	Applicable DID No.	Options
4.4.2	Calibration system description	DI-R-7064	---
4.4.4	Test plan	DI-T-5204	---
4.4.5	Reports, test	DI-T-2072	---

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DOD 5010.12-L, AMSDL. Copies of data item descriptions required by the contractors in connection with specified acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.2.3 Inspection lot formation. For economical purposes, the contracting activity should strongly consider including the acquisition of dummy and training fuzes, in accordance with MIL-D-85580, in with an acquisition for tactical fuzes in accordance with this specification. A concomitant acquisition will eliminate the costlines of a separate acquisition and will greatly facilitate the inspection of all fuzes.

6.3 Definitions.

6.3.1 Safe condition or safe position. Visually, the fuze is considered to be in the safe condition or safe position when a white letter "S" on a green background is clearly visible through the windows located in both ends of the fuze. Electrically, the fuze is considered to be in the safe condition or safe position when the fuze does not initiate the detonators or squibs when 50 ±5 Vdc is applied across pins L(+) and c(-) of connector 6J1, 28 ±4 Vdc is applied across pins J(+) and M(-) of connector 9J1, and 28 ±4 Vdc is momentarily (not less than 100 ms) applied across pins C(+) and B(-) of connector 9J1.

6.3.2 Armed condition or armed position. Visually, the fuze is considered to be in the armed condition or armed position when a white or black letter "A" on a red background is clearly visible through the windows located in both ends of the fuze. Electrically, the fuze is considered to be in the armed condition or armed position when the measured resistance between pins L and c of connector 6J1 is less than 6.0 ohms, or the measured resistance between pins J and M of connector 9J1 is less than 3.9 ohms when 28 ±4 Vdc is applied across pins C(+) and B(-) of connector 9J1.

6.3.3 Safe. Safe is defined as the ability of the fuze to maintain its safety features in a condition which will not create a hazard for personnel or cause any subsequent action to occur during its service life which will nullify the required safety conditions of service use. For nonservice use, safe is defined as the ability of the fuze to maintain its safety features in a condition which will permit disposal without injury to personnel.

6.3.4 Operable. Operable is defined as the ability of the fuze, with specified inputs, to perform to completion its specified function and sequence, and produce the specified outputs within the required operational period or at the specifed point of time. For the environmental test specified herein, operable is further defined as the ability of the fuze to meet the requirements of 3.2.1.1.1 through 3.2.1.6.3 after exposure to any environment.

6.4 Explosives safety precautions. Minimum explosives safety precautions for use by the contractor are detailed in DOD Instruction 4145.26M, DOD Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material.

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6.5 First article. When a first article inspection is required, the item(s) should be a first article sample. The first article should consist of 30 fuzes. The contracting officer should include specific instructions in acquisiton documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.6 Subject term (key word) listing.

Fuze (ordnance)

Ignition safety mechanism

Pyrotechnic device

Safety and arming device

6.7 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to previous issue due to the extensiveness of the changes.

Preparing activity:
Navy-AS

(Project 1336-N571)

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TABLE I. Resistance, voltage, and capacitance requirements at ambient temperature (safe condition).

Item	Fuze Connector test Points (pins)				Limits			Characteristics Tested
	6J1		9J1					
	FROM (+)	TO (-)	FROM (+)	TO (-)	Min	Max	Unit	
1	ALL	CASE	ALL	CASE	10.0	∞	Mohms	Insulation resistance
2	A&C	c&M	-	-	85	115	kohms	R2 1/
3	L	c	-	-	85	115	kohms	R21 S1G & S1F open 2/
4	T	c	-	-	240	350	ohms	ITL solenoid coil resistance
5	T	c	-	-	35	45	kohms	K1 contacts open
6	T	Y	-	-	240	350	ohms	ATI solenoid coil resistance
7	-	-	H	D	0.9	1.5	kohms	R27 parallel K1 coil resistance
8	-	-	E	D	0.9	1.5	kohms	R28 parallel K1 coil resistance
9	-	F	H	-	1.6	1.9	kohms	R24
10	L	M	-	-	0	0.2	ohms	Common wire L&M
11	b	c	-	-	0	0.2	ohms	Common wire b&c
12	W	c	-	-	0	0.2	ohms	Common wire W&c
13	-	-	J	T	0	0.2	ohms	Common wire J&T
14	-	-	K	M	0	0.2	ohms	Common wire K&M
15	-	-	F	D	0	0.2	ohms	Common wire F&D
16	-	-	E	G	0	0.2	ohms	Common wire E&G
17	-	-	H	R	0	0.2	ohms	Common wire H&R
18	H	-	-	D	0	0.2	ohms	Common wire H&D
19	-	-	C	N	0	0.2	ohms	Common wire C&N
20	-	-	P	B	0	0.2	ohms	Common wire P&B
21	Z	c	-	-	0	2.5	ohms	S1B resistance
22	Z	T	-	-	0.6	1.3	volts	Forward voltage CR9
23	Y	T	-	-	0.6	1.3	volts	Forward voltage CR7
24	C	c	-	-	52.0	62.0	volts	Zener voltage VR10
25	A	b	-	-	3/ 0.077	3/ 0.105	Vdc	R2
26	C	M	-	-	8.0	14.0	µF	Capacitance C6

NOTES: 1/ Allow time for circuit to stabilize before making reading.
2/ Connect 1.0 ±0.1 kohm, 5W resistor across pins C and M of connector 6J1 while making measurement.
3/ Multiply limit x V_Z of VR10.

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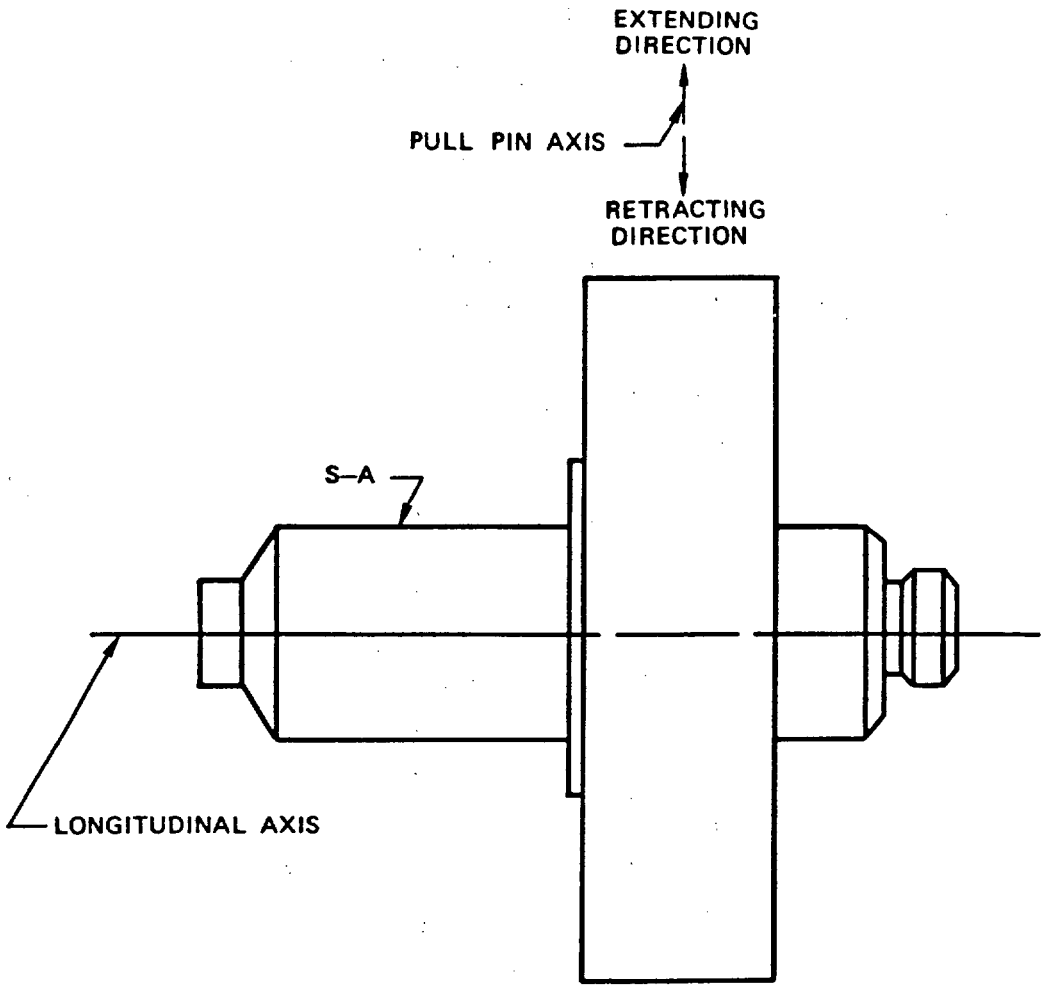
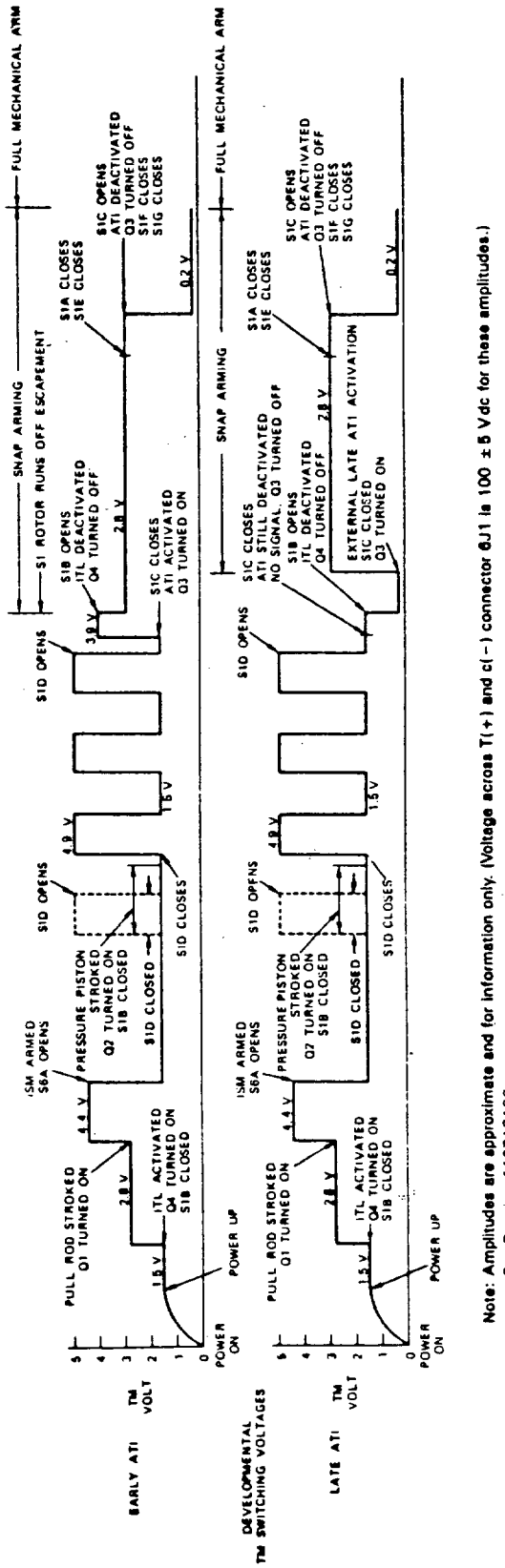


FIGURE 1. Pull pin axis and longitudinal axes.

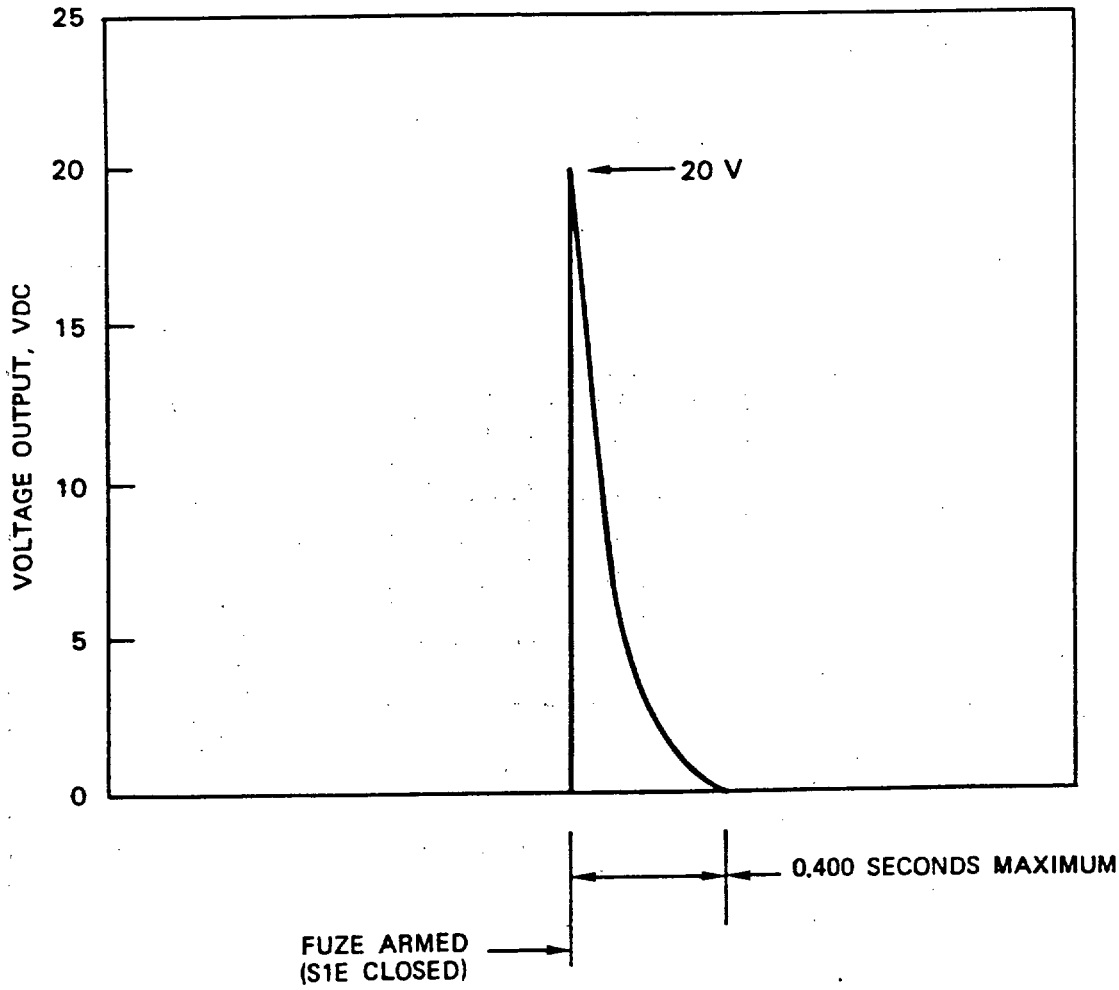
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Note: Amplitudes are approximate and for information only. (Voltage across T(+) and C(-) connector 6J1 is 100 ± 5 Vdc for these amplitudes.)
See Drawing 1137AS196.

FIGURE 2. Electrical arming sequence.

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Note: Voltage is measured across pins R(+) and c(−) of connector 6J1.

FIGURE 3. Fuze tactical TM voltage pulse at armed condition.

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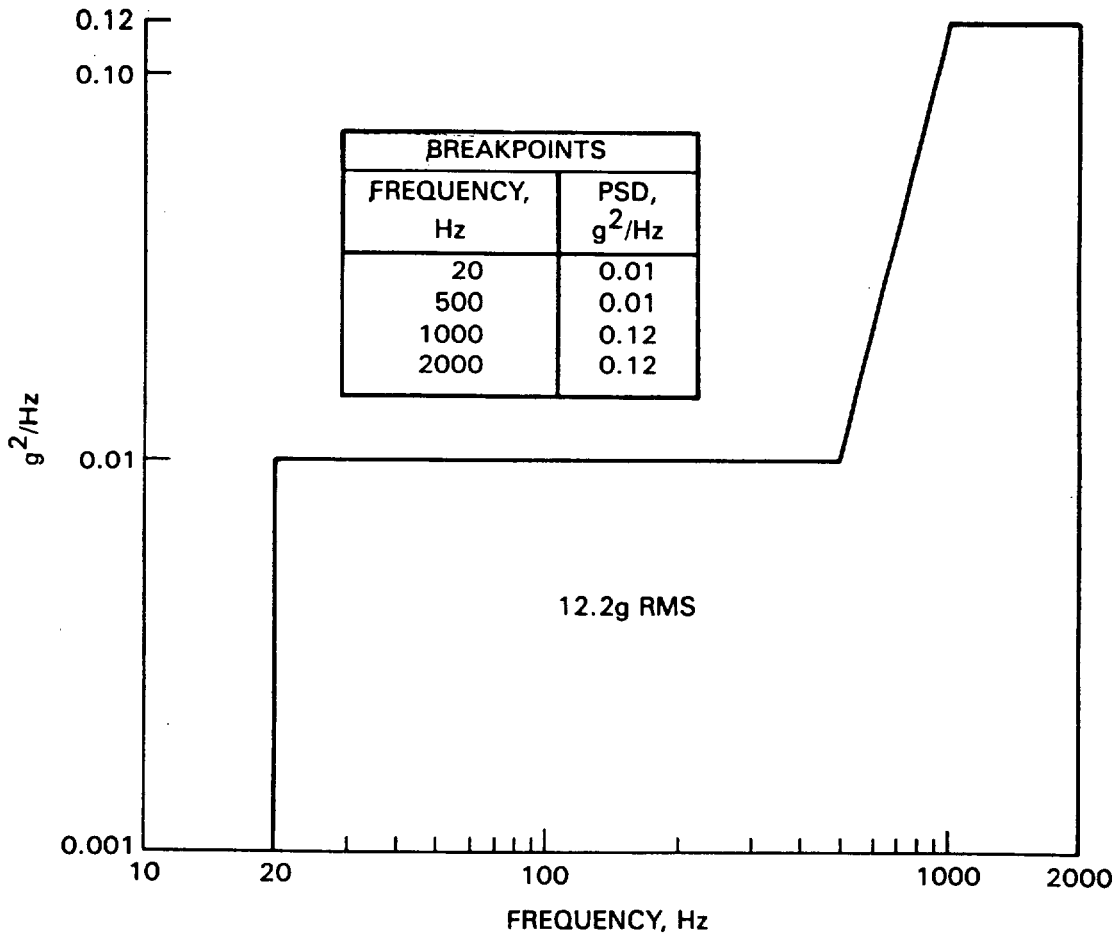


FIGURE 4. Aircraft flight vibration curve.

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BREAKPOINTS	
FREQUENCY, Hz	PSD, g^2/Hz
20	0.02
2000	0.02

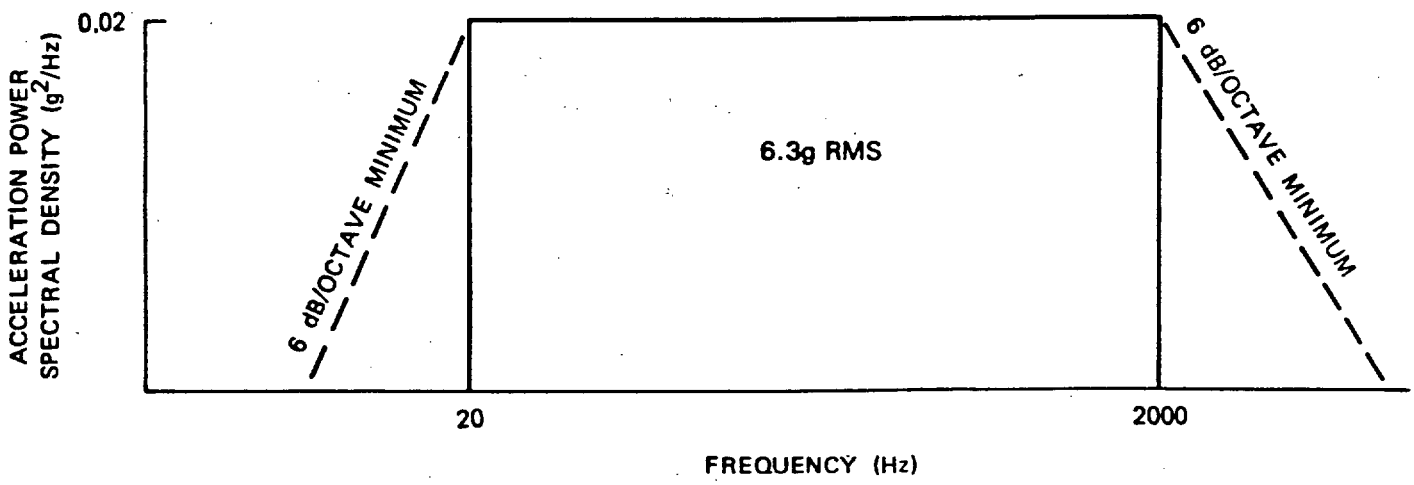
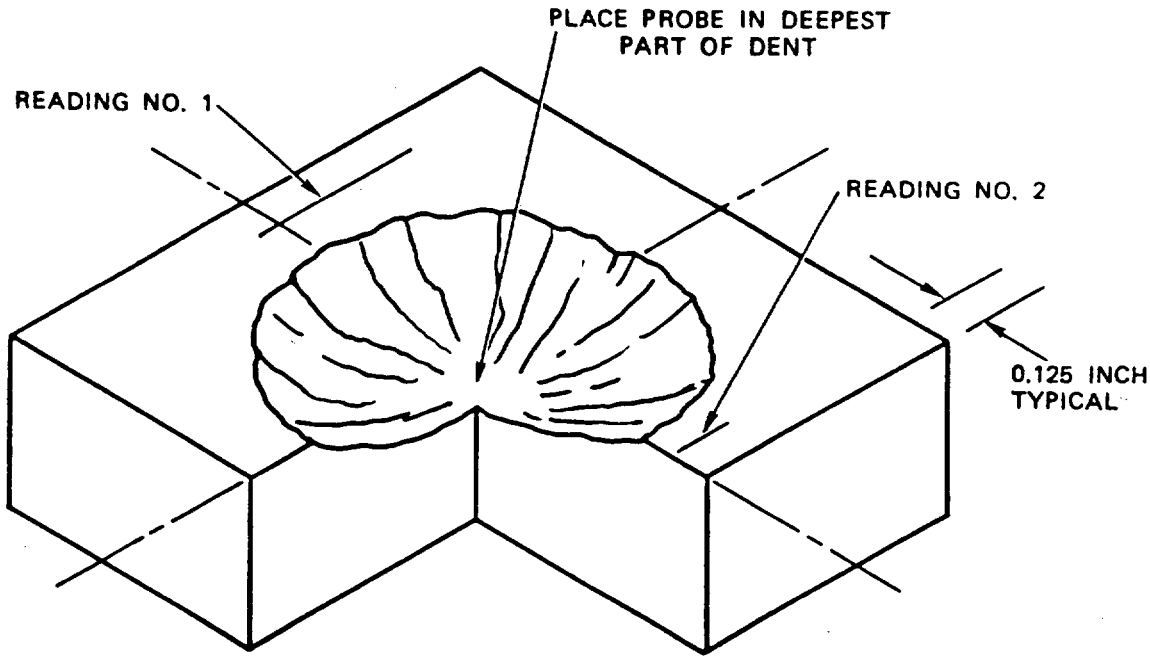


FIGURE 5. Vibration spectral density.

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Note:

DENT BLOCK: Steel, cold finished bar in accordance with ASTM – A 108 Grade 1018 or 1020, Rockwell, hardness B70 to B95, surface condition as rolled. Size, 5.000 ±0.125 inches square, 1.5 inches stock thickness. Block shall be clean and free from rust and burrs.

FIGURE 6. Explosive train output dent block and dent measurement.

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(Front)

SAFE CONDITION TAG
FSU–10/A SAFETY AND ARMING DEVICE, FUZE

Serial No._____ Lot No._____

This fuze was inspected and found to be in the SAFE condition on the date indicated (white letter S on green background visible in both windows).

Signature: Date:

_____ _____
Installing Personnel

_____ _____
Contractor's Representative

_____ _____
Government Inspector

(Back)

This tag shall be placed in the missile folder when the fuze is installed in the missile.

CAUTION: If, at a depot or any using facility, this fuze is found to be in the armed condition, or if it cannot be determined to be in the safe condition, it shall be inspected and verified as to its condition, either armed or indeterminate, by an ordnance officer, and disposed of in accordance with existing regulations for explosive material. A report of the incident, including serial number, nomenclature designation, date of inspection, and any known history of environmental exposure shall be forwarded to the Naval Air Systems Command with this tag.

NOTE: Tag shall be in accordance with UU–T–81, Type B, Grade 15SU, Size 7 or 8, Color white. The required data may be printed or stamped thereon.

FIGURE 7. Sample safe condition tag.

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Notes:

1. Legend:
   - A = Nine fuzes at −65 ± 5°F.
   - B = Eight fuzes at +50 to +104°F.
   - C = Nine fuzes at +160 ±5°F
   - D = Seven fuzes at −65 ±5°F
   - E = Test sequence bracketed by (E) on Figure 9.
   - F = Six fuzes at +50°F to +104°F.
   - G = Seven fuzes at +160 ±5°F.
   - R = Test sequence bracketed by (R) on Figure 9.
   - V = Visual inspection, 4.5.13.
2. All test data shall be recorded.

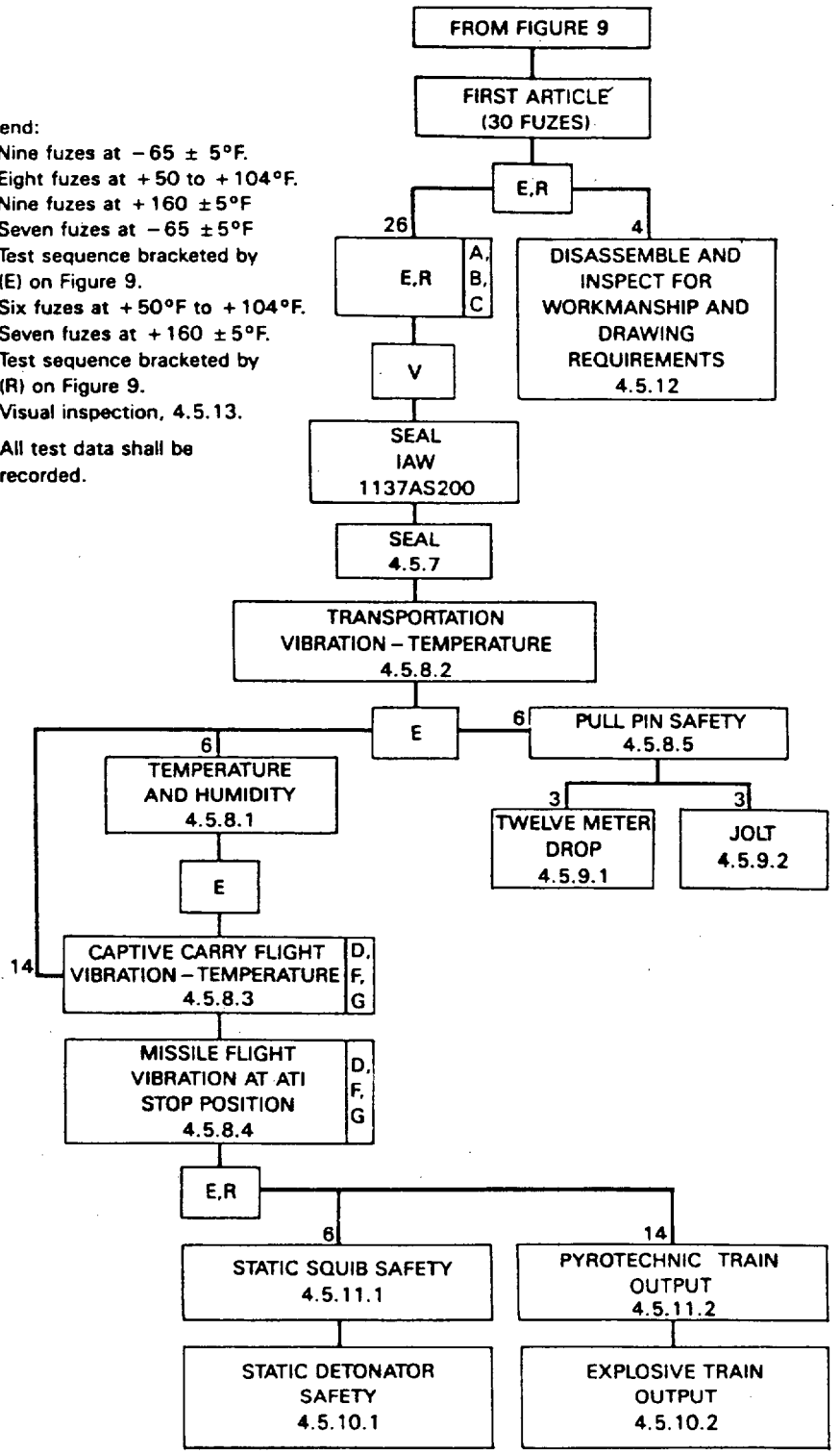
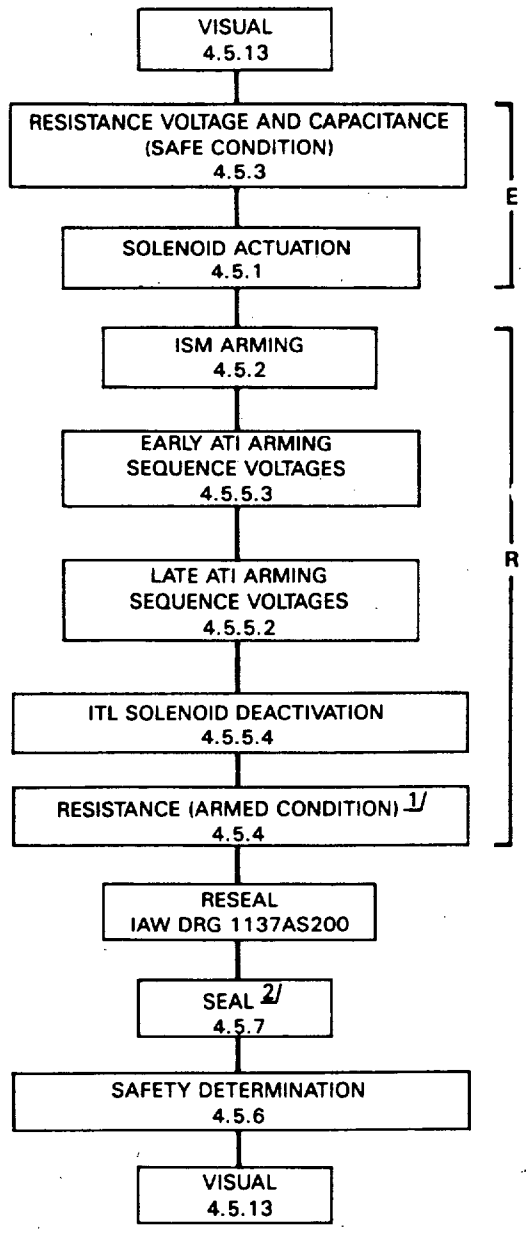


FIGURE 8. First article inspection sequence.

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1/ After test, the fuze shall be recycled to the safe position in accordance with 4.5.6.1.

2/ S – A seal test can be performed prior to, during, or at the end of the performance test. ISM seal test shall be performed at the end of the performance test.

FIGURE 9. Individual unit acceptance inspection sequence.

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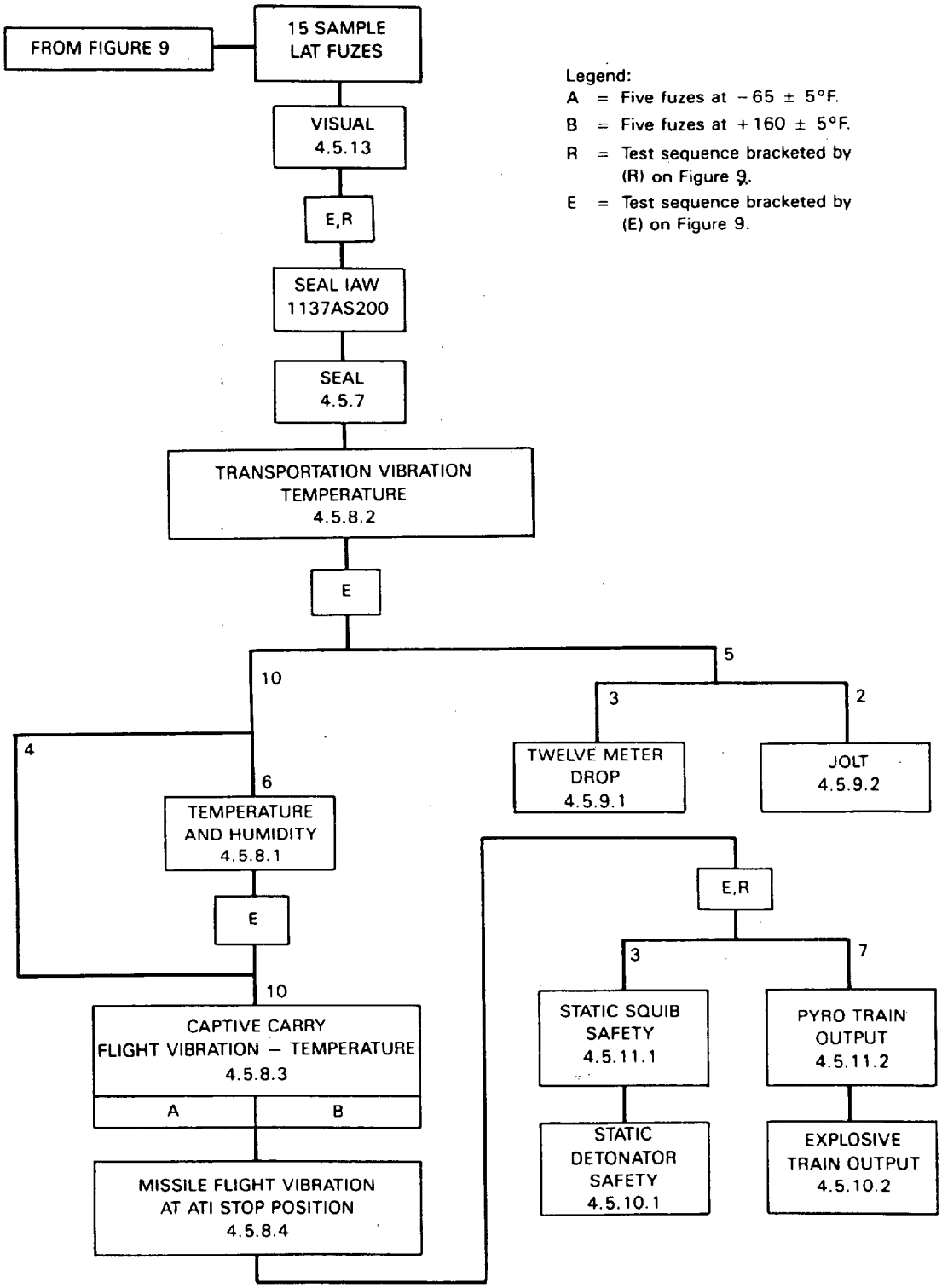


FIGURE 10. Lot acceptance test sample inspection sequence.

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INSPECTION RECORD

FSU – 10/A SAFETY AND ARMING DEVICE, FUZE

Contract No. _____ Contractor _____

First Article ☐

Inspection Lot ☐ S/N _____ Lot No. _____

Rework ☐ Inspector _____ Date _____

DRAWING CONFORMANCE

	Yes	No	Inspector
(1) Two Mk 71 Detonators	☐	☐	_____
(2) Two Mk 24 Leads	☐	☐	_____
(3) One Mk 28 Lead	☐	☐	_____
(4) One Mk 29 Lead	☐	☐	_____
	Pass	Fail	
(5) Drawing 1137AS200	☐	☐	_____

ISM ARMING:

(1) Pin J to M _____ Ohms

(2) Arming time _____ ms

RESISTANCE, VOLTAGE AND CAPACITANCE (SAFE)

(1) All pins to case	_____	Mohms
(2) Pins A & C to c & M	_____	kohms
(3) Pin A to b	_____	Vdc
(4) Pin L to c	_____	kohms
(5) Pin T to c	_____	ohms
(6) Pin T to c	_____	kohms
(7) Pin T to Y	_____	ohms
(8) Pin H to D	_____	kohms
(9) Pin E to D	_____	kohms
(10) Pin F to H	_____	kohms
(11) Pin L to M	_____	ohms
(12) Pin b to c	_____	ohms
(13) Pin W to c	_____	ohms
(14) Pin J to T	_____	ohms
(15) Pin K to M	_____	ohms
(16) Pin F to D	_____	ohms
(17) Pin E to G	_____	ohms
(18) Pin H to R	_____	ohms
(19) Pin H to D	_____	ohms
(20) Pin C to N	_____	ohms
(21) Pin P to B	_____	ohms
(22) Pin Z to c	_____	ohms
(23) Pin Z to T	_____	Vdc
(24) Pin Y to T	_____	Vdc
(25) Pin C to c	_____	V
(26) Pin C to M	_____	μF

	Yes	No
Early ATI Arming Sequence	☐	☐
Late ATI Arming Sequence	☐	☐

S – A ARMING:

EARLY ATI

ARMING TIME _____ SEC

RESISTANCE (ARMED):

Pin L to c _____ ohms

SAFETY DETERMINATION:

Pass Fail Inspector _____

SEAL:

_____ std atm cm^3/s Inspector _____

VISUAL INSPECTION:

	YES	NO
(1) External damage.	☐	☐
(2) Band of yellow adhesive tape present (squibs, detonators, leads); wire seal attached.	☐	☐
(3) Fuze in safe condition. (White "S" visible through both windows.)	☐	☐
(4) Safe condition tag attached, properly filled in, and all signatures present.	☐	☐

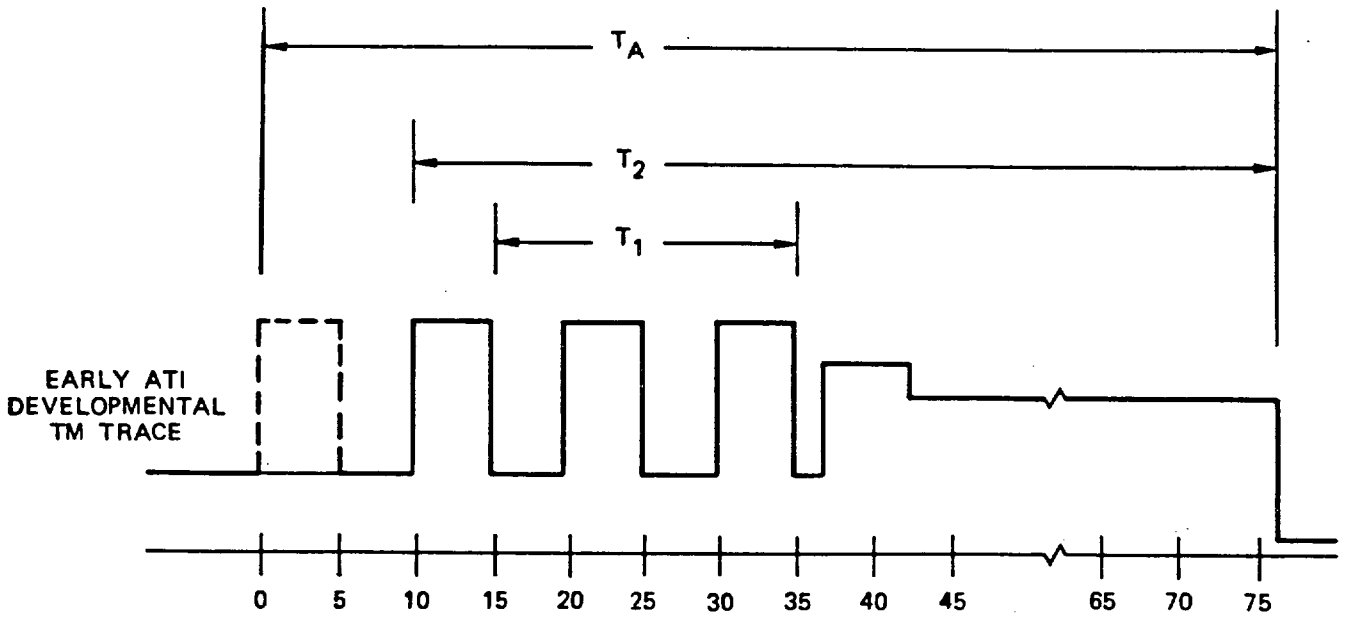
Government representative:

_____ _____

Signature Date

FIGURE 11. Sample fuze inspection record.

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NOTES:

1. LEGEND:

   T_A = S–A arming time

   T_1 = Escapement run time for 20 degrees

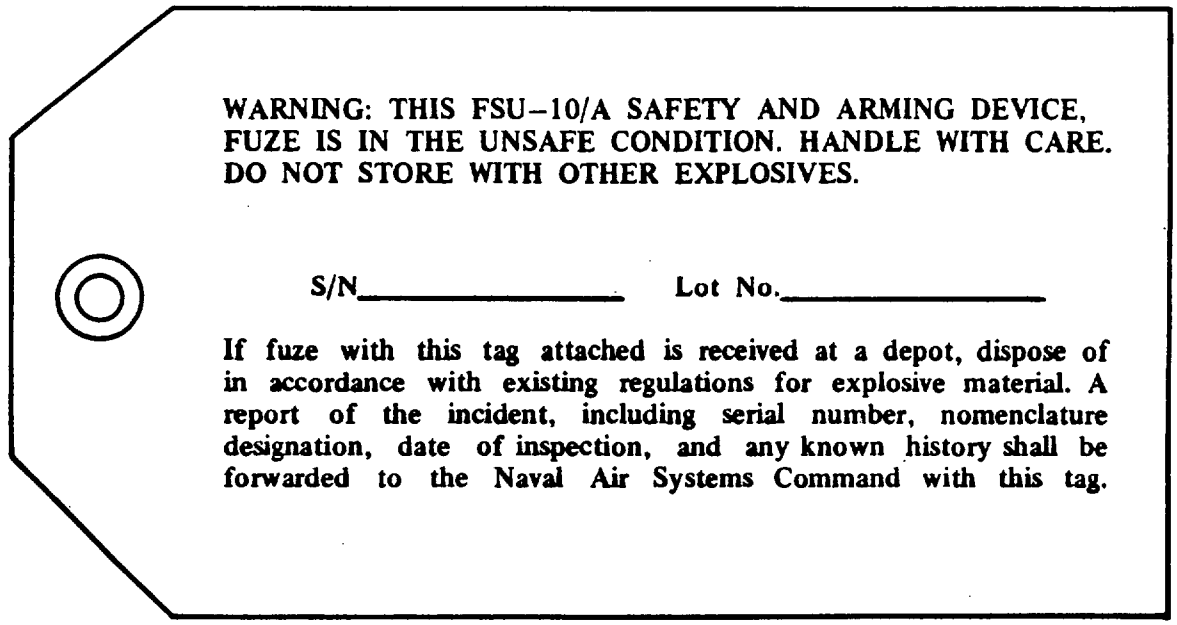
   T_2 = S–A arming time minus escapement run time for 10 degrees

2. S–A arming time equation

   $T_A = 1/2\ T_1 + T_2$

FIGURE 12. Method for calculating S-A arming time.

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NOTE: Tag shall be in accordance with UU-T-81, Type B, Grade 15SU, Size 7 or 8, Color red. The required data may be printed or stamped thereon.

FIGURE 13. Sample unsafe condition tag.

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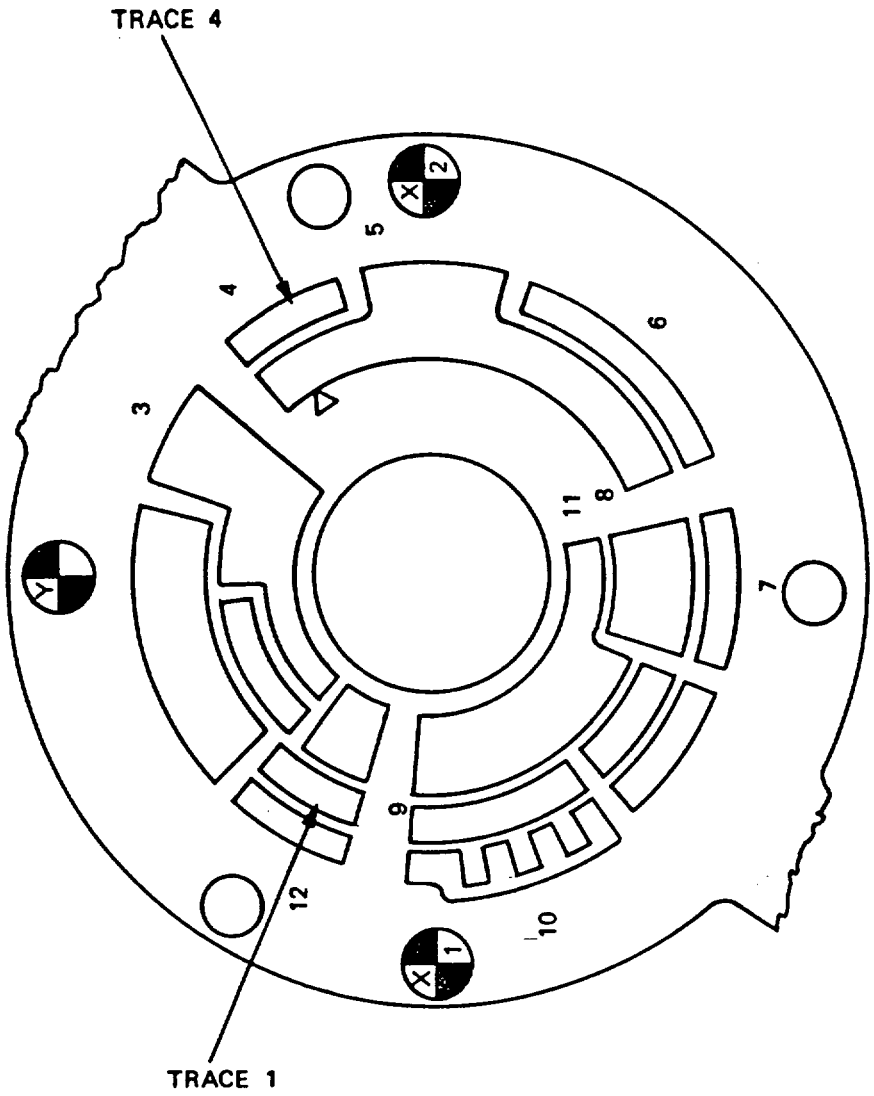


FIGURE 14. Wire attachment to safe-arm stator switch assembly for static detonator safety test.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-85465B(AS)	DEVICE, SAFETY AND ARMING, FUZE FSU-10/A

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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